

# Introduction

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# Teaching Staffs

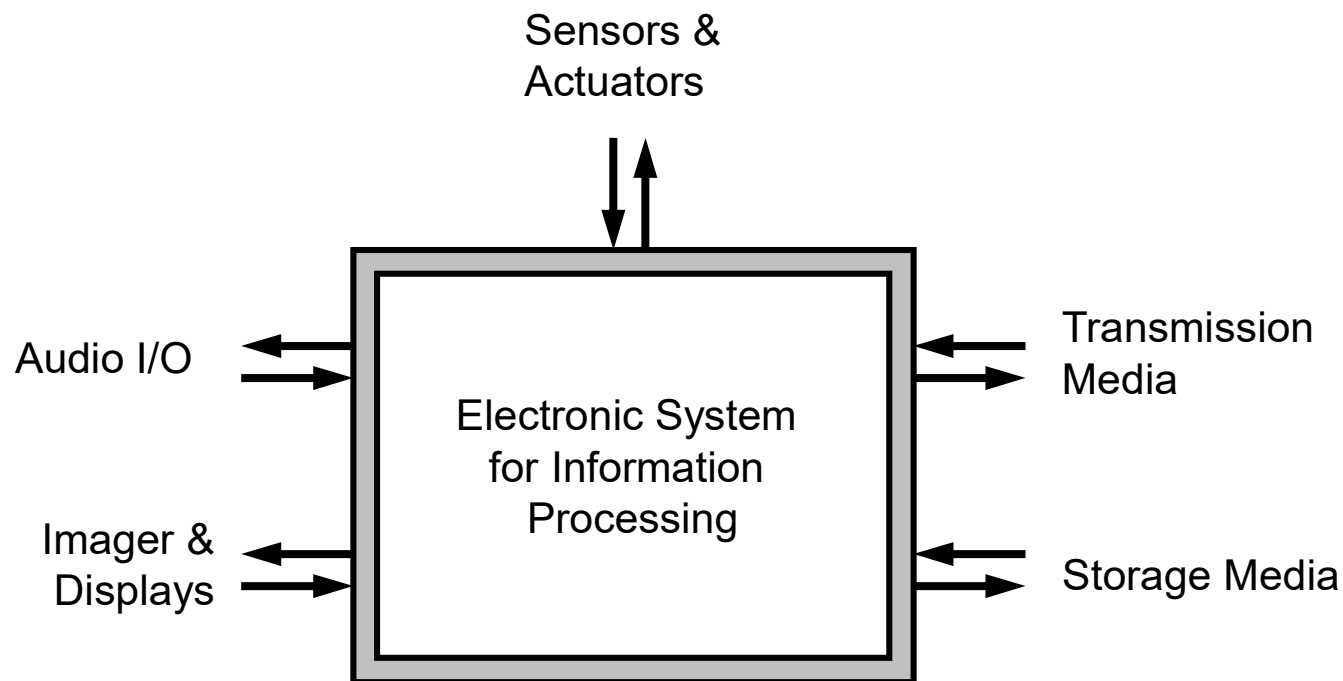
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# Motivation (1)

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- Information is increasingly being processed in the digital domain; yet, the real world is analog by nature
  - Whenever an electronic device wants to talk to the real world, it needs a data converter.



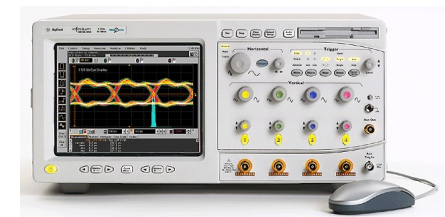
# Motivation (2)

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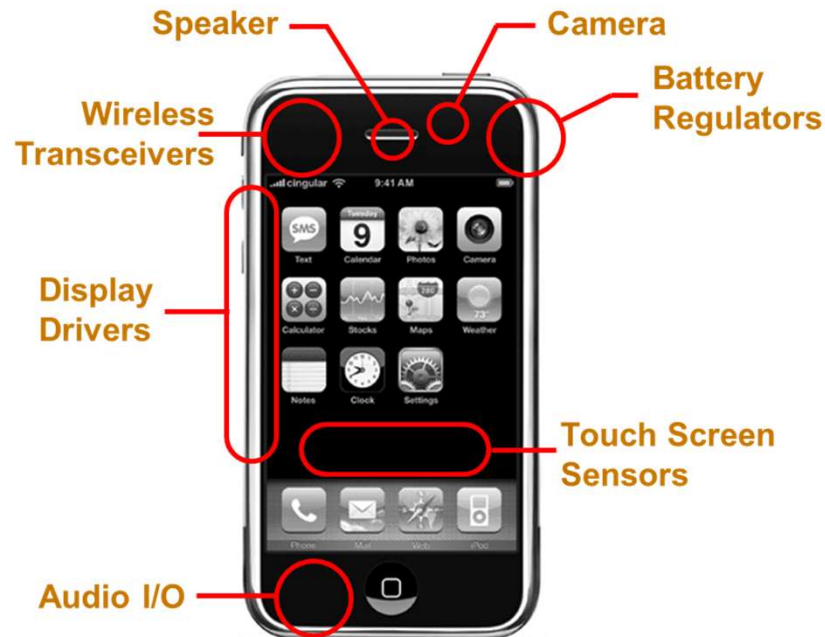
- Benefits of digital signal processing
  - Amenable to automated design and test
  - Enhanced functionality and flexibility
  - Direct benefit from the scaling of VLSI technology
  - Reduced sensitivity to "analog" noise
  - "Arbitrary" precision
- Issues
  - Data converters are difficult to design
    - Especially due to ever-increasing performance requirements
  - Data converters often present a performance bottleneck
    - Speed, resolution or power dissipation of the A/D or D/A converter can limit overall system performance

# Data Converter Applications

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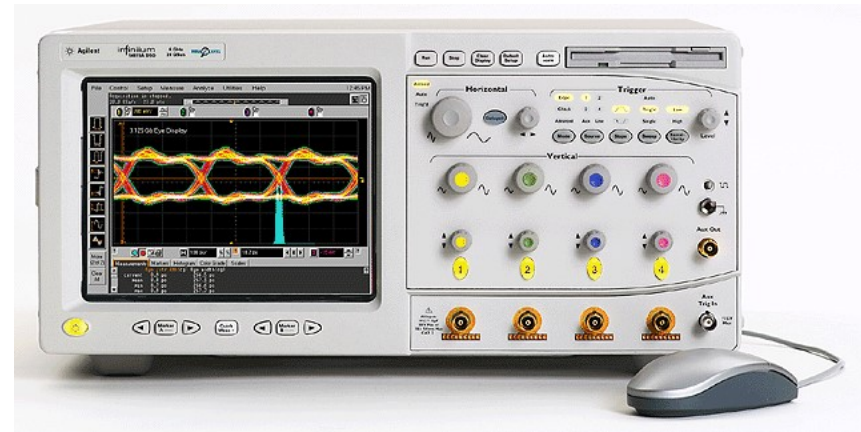


# Examples



More than 20 data converters are used in each iPhone system

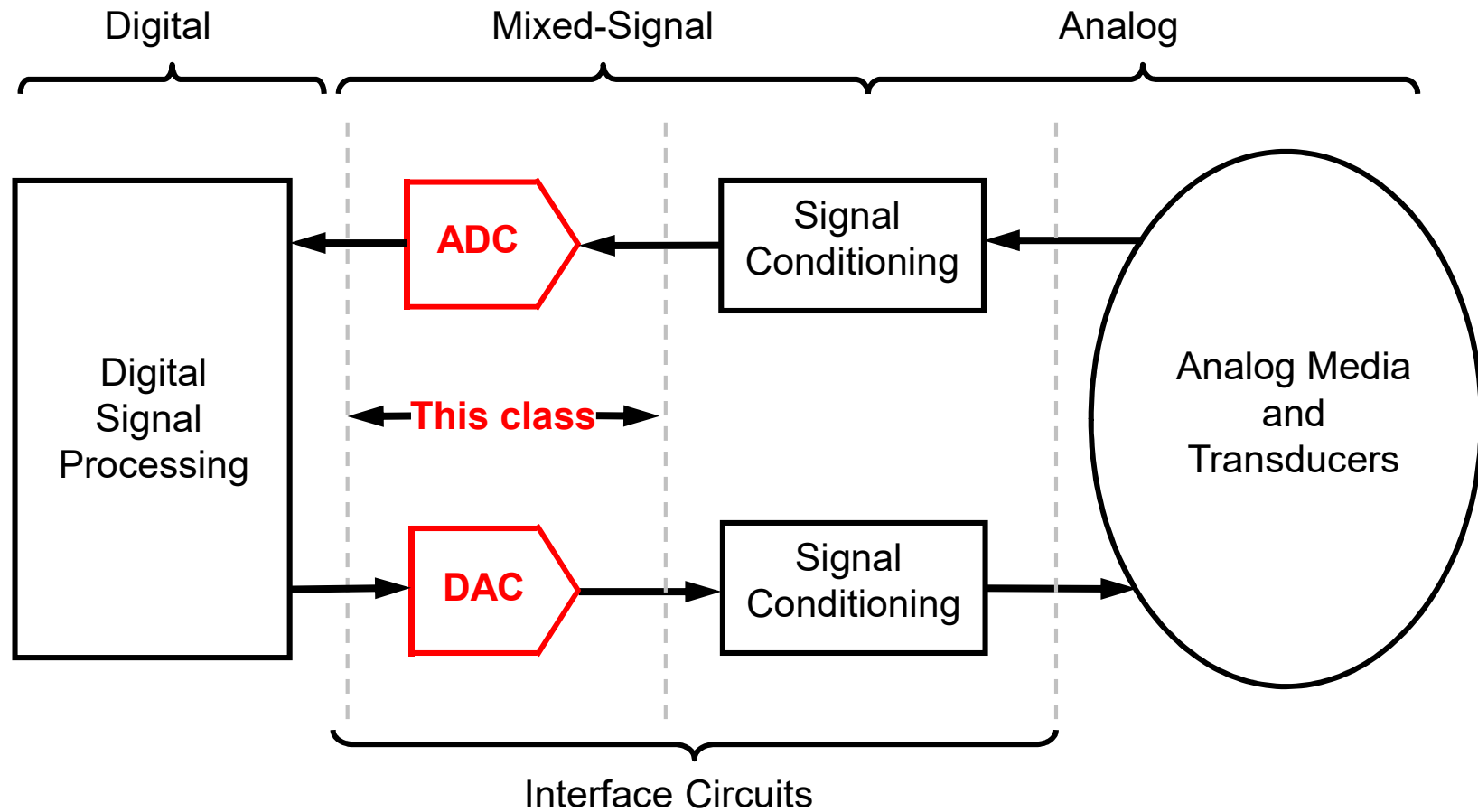
- A typical cell phone contains more than 20 data converter



- Digital oscilloscopes are essentially high-performance ADCs

# Course Contents

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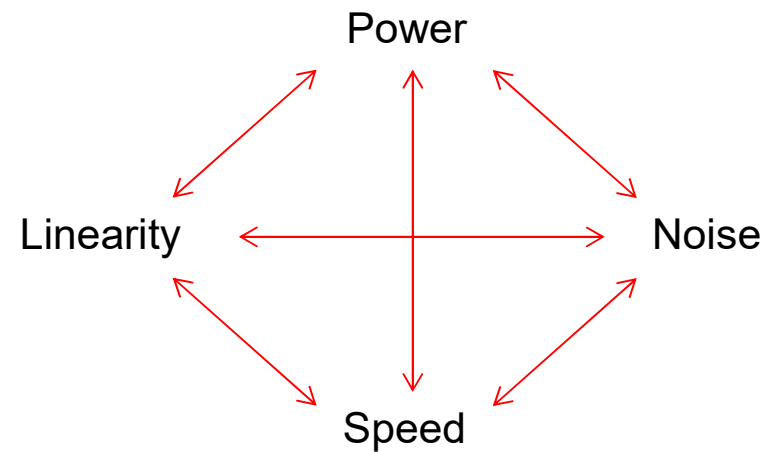


# How I look at data converters

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Crown in analog mixed-signal IC design  
Difficult but fun!



# Course Prerequisites

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- Undergraduate electronic circuit courses:
  - Device physics and models
  - Transistor level analog circuits
  - Frequency response, feedback, and noise
- Graduate circuit course:
  - Analog IC design
- Prior exposure to SPICE (Cadence) and Matlab

# Course Topics

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- Ideal sampling, reconstruction, & quantization
- Sampling circuits
- Voltage comparators
- Nyquist-rate ADCs and DACs
- Oversampled ADCs and DACs
- Data converter performance trends and limits
- Data converter testing